The Reynolds Transport Theorem (RTT) & Control Volume Selection Strategy

Converting Lagrangian material derivatives into Eulerian flux balances across fixed, moving, and deformable control boundaries.

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1. The Reynolds Transport Theorem Master Formula

In continuum mechanics, physical conservation laws (conservation of mass, Newton's second law, and the first law of thermodynamics) are fundamentally formulated for a **closed material system**: a fixed collection of fluid particles enclosed within a time-dependent, deforming material volume $V_f(t)$ bounded by surface $\Sigma_f(t)$ that moves along with the local fluid velocity v.

Let B_{sys} denote any extensive property of this closed system (such as mass m, linear momentum mv, angular momentum H_{x_0}, or total energy E), and let $b = dB/dm$ be the corresponding intensive property per unit mass. The total extensive property stored in the fluid volume is:

$$B_{\text{sys}}(t) = \int_{V_f(t)} \rho b \, dV$$

[Eq. 3.1] Total extensive property B_{sys} as the volume integral of volumetric density $\phi = \rho b$.

To differentiate this integral with respect to time when both the integrand ρb and the spatial domain $V_f(t)$ vary simultaneously, we apply the formal analytical limit definition:

$$\frac{dB_{\text{sys}}}{dt} = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \left[\int_{V_f(t+\Delta t)} \rho b(x, t+\Delta t) \, dV - \int_{V_f(t)} \rho b(x,t) \, dV \right]$$

[Eq. 3.2] Analytical limit definition of the material time derivative.

Over the infinitesimal time interval Δt, the material boundary moves with fluid velocity v. An elementary surface patch $d\sigma$ sweeps out a cylindrical volume of base $d\sigma$ and height $(v\Delta t) \cdot n$, yielding $dV = (v \cdot n)\Delta t \, d\sigma$. Splitting the limit into local unsteadiness and boundary convective sweeping yields the classical RTT for a material volume:

$$\frac{d}{dt}\left[\int_{V_f(t)} \rho b \, dV\right] = \int_{V_f(t)} \frac{\partial(\rho b)}{\partial t} \, dV + \int_{\Sigma_f(t)} \rho b (v \cdot n) \, d\sigma$$

[Eq. 3.6] Reynolds Transport Theorem for a material fluid volume $V_f(t)$.

Universal Extension to an Arbitrary Moving Control Volume $V_c(t)$

In aerospace propulsion and aerodynamics, tracking material fluid packages downstream of a jet engine is impossible. Instead, we define a geometric **control volume** $V_c(t)$ whose control surface $\Sigma_c(t)$ moves with an arbitrary, specified velocity field $v_c(x,t)$. Matching $V_c(t)$ with $V_f(t)$ instantaneously at time t yields the **Universal RTT Master Formula**:

$$\mathbf{\frac{dB_{sys}}{dt} = \frac{d}{dt}\left[\int_{V_c(t)} \rho b \, dV\right] + \int_{\Sigma_c(t)} \rho b [(v - v_c) \cdot n] \, d\sigma}$$

[Eq. 3.8] Universal Reynolds Transport Theorem with relative velocity $(v - v_c)$.

> PHYSICAL INTERPRETATION OF THE TWO TERMS
>
> **1. Rate of Storage / Unsteadiness:** $\frac{d}{dt}\int_{V_c} \rho b \, dV$ measures how much of property B accumulates or depletes inside the instantaneous geometric boundaries of $V_c(t)$.
>
> **2. Net Convective Flux:** $\int_{\Sigma_c} \rho b [(v - v_c) \cdot n] \, d\sigma$ accounts for property B carried across the boundary by the *relative velocity* $w = v - v_c$ with which fluid crosses the control surface.

Step-by-Step Derivation of the Reynolds Transport Theorem (Notes.pdf, Eqs. 3.2-3.8)

This section expands every step that the compact derivation above skips. Throughout, $\phi(x,t)$ is the volumetric density of the extensive property (in the notation $B = \int \rho b \, dV$ used elsewhere, $\phi = \rho b$). It is assumed twice continuously differentiable in space and time, and the surfaces $\Sigma_f(t)$ and $\Sigma_c(t)$ are smooth and closed, with n the outward unit normal. Equation numbers are those of Notes.pdf, Chapter 3.

Step 1: Split the difference quotient (Eqs. 3.2 and 3.3)

The material volume at $t + dt$ is the volume at t plus the thin shell $\delta V = V_f(t+dt) - V_f(t)$ swept by its boundary. The shell is counted algebraically: where the boundary moves inward it removes volume. Add and subtract $\int_{V_f(t)} \phi(x, t+dt) \, dV$ in the numerator of the limit definition (Eq. 3.2):

$$\int_{V_f(t+dt)} \phi(x,t+dt)\,dV - \int_{V_f(t)} \phi(x,t)\,dV = \int_{V_f(t)} \big[\phi(x,t+dt) - \phi(x,t)\big]\,dV + \int_{\delta V} \phi(x,t+dt)\,dV$$

Eq. 3.2 rewritten as Eq. 3.3 (before the limit)

The first integral has a fixed domain $V_f(t)$ and measures how the field changes at fixed points. The second integral contains the field only over the thin shell and measures how much the domain grows.

Step 2: The unsteadiness term (Eq. 3.4)

Taylor-expand the integrand at a fixed point x:

$$\phi(x,t+dt) - \phi(x,t) = \frac{\partial \phi}{\partial t}\,dt + O(dt^2)$$

Because ϕ is C^2 and $V_f(t)$ is bounded, the remainder is uniformly $O(dt^2)$ over the domain, so it can be integrated. Dividing by dt and letting $dt \to 0$ (the domain $V_f(t)$ does not depend on dt):

$$\lim_{dt\to 0} \frac{1}{dt} \int_{V_f(t)} \big[\phi(x,t+dt) - \phi(x,t)\big]\,dV = \int_{V_f(t)} \frac{\partial \phi}{\partial t}\,dV$$

Eq. 3.4

Step 3: The swept-volume term (Eq. 3.5)

A point x_s of the material surface moves in the time dt to $x_s + v\,dt + O(dt^2)$. Its displacement along the outward normal is $(v \cdot n)\,dt$. The tangential part only slides the point along the surface and adds no volume. The shell above a surface element $d\sigma$ is therefore a prism of base $d\sigma$ and height $(v \cdot n)\,dt$ (an oblique prism has the same volume as a right prism of equal base and height). Where $v \cdot n < 0$ the height is negative and the shell removes volume, which is the algebraic counting adopted in Step 1:

$$d(\delta V) = (v \cdot n)\,dt\,d\sigma + O(dt^2)$$

Since $\phi(x,t+dt) = \phi(x,t) + O(dt)$, the integrand can be evaluated at time t at the cost of an $O(dt^2)$ error:

$$\int_{\delta V} \phi(x,t+dt)\,dV = dt \int_{\Sigma_f(t)} \phi(x,t)\,v \cdot n\,d\sigma + O(dt^2)$$

Dividing by dt and taking the limit gives the convective flux through the moving boundary:

$$\lim_{dt\to 0}\frac{1}{dt}\int_{\delta V}\phi(x,t+dt)\,dV=\int_{\Sigma_f(t)}\phi\,v\cdot n\,d\sigma$$

Eq. 3.5

Step 4: Add both contributions (Eq. 3.6)

The limit of the sum is the sum of the limits, so Eqs. 3.4 and 3.5 give Eq. 3.6:

$$\frac{d}{dt}\int_{V_f(t)}\phi\,dV=\int_{V_f(t)}\frac{\partial\phi}{\partial t}\,dV+\int_{\Sigma_f(t)}\phi\,v\cdot n\,d\sigma$$

Eq. 3.6

Step 5: Arbitrary moving control volume (Eq. 3.7)

A control volume $V_c(t)$ is a geometric region whose boundary points move with a prescribed velocity $v_c(x,t)$, not necessarily equal to the fluid velocity. Steps 1 to 4 used the velocity of the boundary only to compute the displacement $(v\cdot n)\,dt$ of the surface. Repeating them with the boundary velocity v_c in place of v gives

$$\frac{d}{dt}\int_{V_c(t)}\phi\,dV=\int_{V_c(t)}\frac{\partial\phi}{\partial t}\,dV+\int_{\Sigma_c(t)}\phi\,v_c\cdot n\,d\sigma$$

Eq. 3.7

Step 6: Subtraction at the instant of coincidence (Eq. 3.8)

Choose $V_c(t)$ so that at the instant t it occupies exactly the same region as $V_f(t)$, hence $\Sigma_c(t)=\Sigma_f(t)$ and the normal n is the same. (At later times the two volumes differ, which is why only the instantaneous values of the derivatives are compared.) The integrals of $\partial\phi/\partial t$ in Eqs. 3.6 and 3.7 are then identical. Subtracting Eq. 3.7 from Eq. 3.6:

$$\frac{d}{dt}\int_{V_f}\phi\,dV-\frac{d}{dt}\int_{V_c}\phi\,dV=\int_{\Sigma_c}\phi\,v\cdot n\,d\sigma-\int_{\Sigma_c}\phi\,v_c\cdot n\,d\sigma$$

Combining the two surface integrals gives Eq. 3.8:

$$\frac{d}{dt}\int_{V_f(t)} \phi \, dV = \frac{d}{dt}\int_{V_c(t)} \phi \, dV + \int_{\Sigma_c(t)} \phi \, (v - v_c) \cdot n \, d\sigma$$

Eq. 3.8

Step 7: Checks (extension, not in the Notes)

One-dimensional check with the Leibniz rule. Take a one-dimensional "volume" $a(t) \le x \le b(t)$. The outward normals are $n = +1$ at $x = b$ and $n = -1$ at $x = a$, and the boundary velocities are $\dot{b}$ and $\dot{a}$. Eq. 3.7 gives $\int_a^b \partial\phi/\partial t \, dx + \phi(b,t)\dot{b} - \phi(a,t)\dot{a}$. Differentiating $F(t) = \int_{a(t)}^{b(t)} \phi(x,t)\, dx$ directly with the chain rule for $F(t,a,b)$ and Barrow's rule $\partial F/\partial b = \phi(b,t)$, $\partial F/\partial a = -\phi(a,t)$:

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + \frac{\partial F}{\partial b}\dot{b} + \frac{\partial F}{\partial a}\dot{a} = \int_a^b \frac{\partial \phi}{\partial t}\, dx + \phi(b,t)\,\dot{b} - \phi(a,t)\,\dot{a}$$

Both results coincide, which confirms the sign convention of the flux term.

Dimensional check. If $[\phi]$ denotes the units of the density, then $[d\Phi/dt] = [\phi]\,\mathrm{m}^3/\mathrm{s}$, the unsteady term has units $[\phi]\,\mathrm{m}^3/\mathrm{s}$, and the flux term has units $[\phi] \cdot (\mathrm{m/s}) \cdot \mathrm{m}^2 = [\phi]\,\mathrm{m}^3/\mathrm{s}$.

Limiting case. If $v_c = v$ at the boundary, the relative velocity vanishes and Eq. 3.8 reduces to the identity $d\Phi_f/dt = d\Phi_c/dt$, as expected for a volume that moves with the fluid.

2. The Control Volume Toolbox: Selection Strategy

The entire art of integral fluid mechanics lies in choosing the optimal control volume V_c. The choice of V_c does not change the physical outcome, but a clever choice reduces pages of vector calculus to a 2-line algebraic equation.

Fixed Control Volume

$V_0, v_c = 0$

Boundaries are fixed in the laboratory frame. Convective velocity is simply v. Ideal for wind tunnels, stationary pipe bends, jet nozzles on testbenches, and steady heat exchangers.

Moving Control Volume

$V_c(t), v_c \neq 0$

Translates at constant speed v_c or accelerates with a vehicle (rockets, cruising aircraft, traveling shock waves, hydraulic bores). Fluxes are carried by relative velocity $w = v - v_c$.

Deformable Control Volume

$V_c(t), v_c(x,t)$

Boundary velocity varies along the control surface. Essential for reciprocating internal combustion engine pistons, inflating airbags,

expanding rocket propellant bladders, and flexible vessels.

Boundary Decomposition and Sign Conventions

A control surface Σ_c is always partitioned into functional segments:

$$\Sigma_c = \Sigma_{\text{in}} \cup \Sigma_{\text{out}} \cup \Sigma_{\text{wall}} \cup \Sigma_{\text{free}}$$

- **Outward unit normal n:** By universal convention, n is *strictly outward-pointing* from V_c at every boundary element $d\sigma$.
- **Inlets (Σ_{in}):** Fluid enters the volume against the normal vector: $(v - v_c) \cdot n < 0$. This negative sign automatically ensures that inflow increases internal storage!
- **Outlets (Σ_{out}):** Fluid leaves the volume along the normal vector: $(v - v_c) \cdot n > 0$.
- **Impermeable Walls (Σ_{wall}):** Fluid cannot cross a solid wall. Kinematically, $(v - v_c) \cdot n = 0$, so convective flux through solid walls is identically zero.

3. The Moving Reference Frame Trick

One of the most powerful analytical techniques in aerospace engineering is **converting an unsteady flow problem into a steady one** by attaching the reference frame and control volume to a traveling disturbance.

Consider a solitary wave, a hydraulic bore, or a normal shock wave propagating at velocity $c = c\hat{i}$. In the fixed laboratory frame, the fluid height, pressure, and velocity at any point x change rapidly as the front sweeps past ($\partial/\partial t \neq 0$), requiring complex partial differential equations.

The Trick: Attach a control volume V_c directly to the wave front, moving with constant velocity $v_c = c\hat{i}$. Observed from this moving frame:

1. The wave front is completely stationary inside V_c.
2. The flow is entirely **steady**: $\frac{d}{dt}\int_{V_c} \rho b \, dV \equiv 0$.
3. Fluid enters from the right with relative speed $w_1 = v_1 - c$ and leaves to the left with relative speed $w_2 = v_2 - c$.

All time derivatives vanish, transforming the problem into a simple 1D steady flux balance!

CANONICAL PROBLEM WALKTHROUGH · CL2 SETUP

The Hydraulic Bore / Moving Shock Wave

Problem Statement: A tidal surge creates a hydraulic bore of height h_2 moving with constant wave velocity c into a calm, shallow channel of depth h_1 ($h_2 > h_1$) where water is initially at rest ($v_1 = 0$). Behind the bore, fluid moves forward with induced speed v_b. Determine the bore propagation speed c and induced velocity v_b in terms of h_1, h_2, and g.

Stage 1 Select Moving Control Volume V_c

Attach a 2D control volume of width W to the moving wave front, translating at velocity $v_c = c\hat{i}$. In this frame, the flow is steady ($\partial/\partial t = 0$).

Calculate relative horizontal velocities:

- **Inflow (upstream ahead of bore):** Undisturbed water at rest ($v_1 = 0$) approaches the front with relative speed:

$$w_1 = v_1 - c = -c$$

- **Outflow (downstream behind bore):** Water moving at $v_b\hat{i}$ leaves the front with relative speed:

$$w_2 = v_b - c = -(c - v_b)$$

Stage 2 Apply Mass Conservation (Continuity)

Using the steady 1D continuity equation ($\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}}$):

$$\rho W h_1 |w_1| = \rho W h_2 |w_2| \implies h_1 c = h_2 (c - v_b)$$

Solving for the relative exit speed and induced speed v_b:

$$c - v_b = c\frac{h_1}{h_2} \implies v_b = c\left(1 - \frac{h_1}{h_2}\right) = c\frac{h_2 - h_1}{h_2}$$

Stage 3 Apply Linear Momentum Conservation

For a steady moving control volume with negligible bed friction over the short bore length:

$$\sum F_{x,\text{ext}} = \dot{m}(w_{2,x} - w_{1,x})$$

External horizontal forces are purely hydrostatic pressure distributions:

- Right face (height h_1): $F_{p1} = \int_0^{h_1} \rho g (h_1 - y) W \, dy = \frac{1}{2}\rho g W h_1^2$ (pushes left, $-\hat{i}$).
- Left face (height h_2): $F_{p2} = \int_0^{h_2} \rho g (h_2 - y) W \, dy = \frac{1}{2}\rho g W h_2^2$ (pushes right, $+\hat{i}$).

Net horizontal force (taking right as $+x$):

$$\sum F_x = \frac{1}{2}\rho g W h_2^2 - \frac{1}{2}\rho g W h_1^2 = \frac{1}{2}\rho g W (h_2^2 - h_1^2)$$

Net convective momentum flux leaving the volume (note $w_{2,x} = -(c - v_b)$ and $w_{1,x} = -c$):

$$\dot{m}(w_{2,x} - w_{1,x}) = (\rho W h_1 c)[-(c - v_b) - (-c)] = \rho W h_1 c v_b$$

Stage 4

Solve for Wave Speed c and Induced Velocity v_b

Equating net force to momentum flux:

$$\frac{1}{2}\rho g W (h_2^2 - h_1^2) = \rho W h_1 c v_b$$

Substitute $v_b = c\frac{h_2 - h_1}{h_2}$ into the right-hand side:

$$\frac{1}{2} g (h_2 - h_1)(h_2 + h_1) = h_1 c^2 \frac{h_2 - h_1}{h_2}$$

Dividing by $(h_2 - h_1) \neq 0$ yields the celebrated **Hydraulic Bore Propagation Speed Formula**:

$$\mathbf{c = \sqrt{g h_2 \frac{h_1 + h_2}{2 h_1}}}$$

And substituting back into v_b:

$$\mathbf{v_b = (h_2 - h_1)\ \frac{g(h_1 + h_2)}{2h_1h_2}}$$

AEROSPACE EXAM INSIGHT: ACOUSTIC LIMIT

In the limit of an infinitesimal wave ($h_2 \to h_1$), $c \to \quad gh_1$, which is the shallow water gravity wave propagation speed (exact hydraulic analog of the acoustic speed of sound $a = \quad \gamma RT$). For finite jumps ($h_2 > h_1$), the bore is supersonic ($Fr = c/ \quad gh_1 > 1$), exactly analogous to a normal shock wave in compressible gas dynamics!

← Topic 3 Hub

Next: 3.2 Conservation of Mass →

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